University of Cape Town
Department of Computer Science

Computer Science CSC3003S
Class Test 1, 2016

Answer all questions.
Approximate marks per question are shown in brackets.

Time: 40 minutes **Marks: 40**

Section 1. ASD [32 Marks]

Question 1. **[12]**

i. There are three roles in SCRUM. Name the roles and describe them and their responsibilities. [9]

== ANSWER ==
One point for the name of the role, up to two points per role for the description
Product Owner [1]

- represents customer interests by managing the product backlog [1]
- manages overall projects *any of the points below or variations* [1]
    - represents the interests of everyone with a stake in the prjoect
    - achieves initial and ongoing funding
    - creates project overall requirements, RoI objectives, release plan

Team [1]

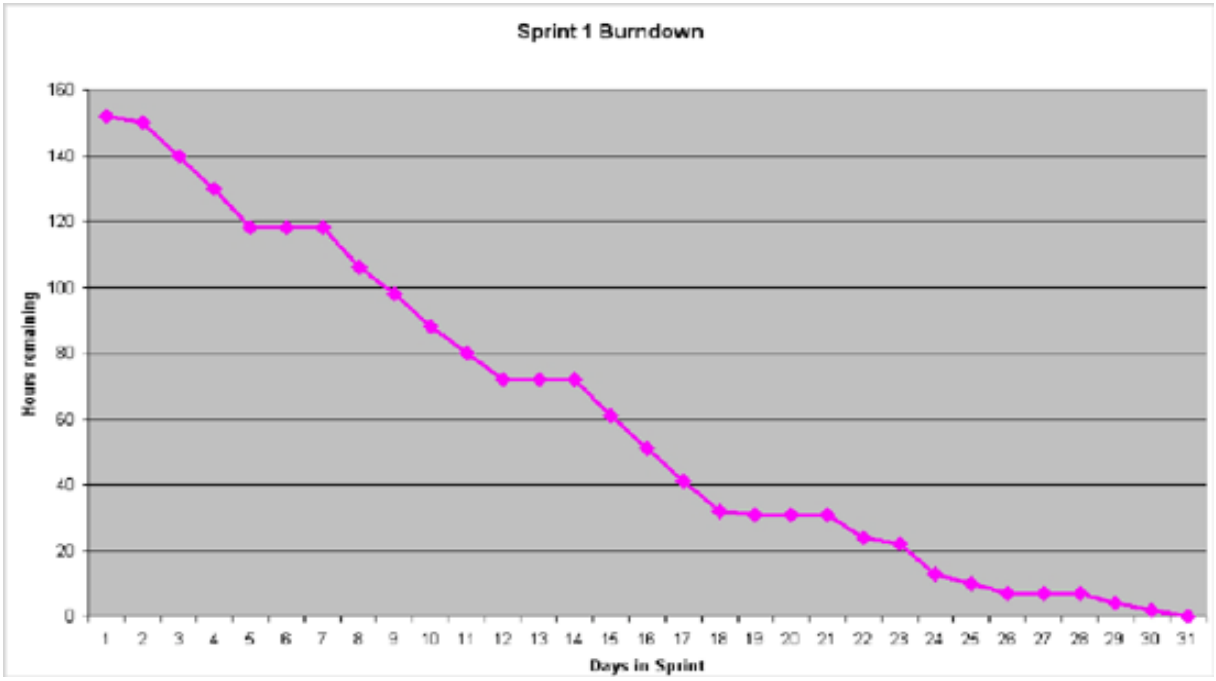
- Half point for each of the following [2]
    - decides how to turn product backlog into an increment of functionality
    - develops functionality
    - responsible for managing own work
    - responsible for the success of each iteration and the project as a whole

Scrum Master [1]

- Teaching scrum to everyone in the project
- Implementing scrum so it fits within the organization's culture and still delivers the expected benefits
- Ensuring that everyone follows scrum rules and practices

ii. Draw an "ideal" SCRUM sprint burndown chart. [3]

== ANSWER ==



Key features:

- X-axis is labelled “days in sprint” and Y-axis is labelled “hours remaining” [1]
- Line starts at the top and ends at 30 days and 0 hours [1]
- In an ideal sprint, no work is done on weekends (i.e. line is flat) [1]
- Line can go upwards at the beginning but should generally be trending downwards

Question 2. **[5]**

i. One of the key aspects of project management is setting the scope for a project. Explain what “scope” entails [3]

== ANSWER ==
Project scope is (one point each, up to three)

- Goals derived from the needs and expectations of the stakeholders
- the functions and features that are to be delivered to end-user
- data that are input and output
- content that is presented to users as a consequence of using the software
- performance, constraints, interfaces, and reliability that bound the system

ii. How might one proceed with defining scope [2]

== ANSWER ==
It might be defined using one of two techniques (either answer is acceptable)

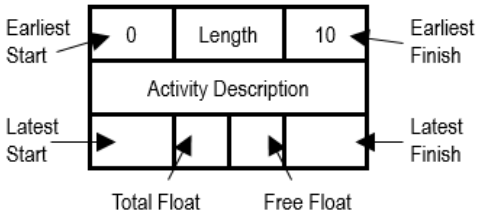
- A narrative description of the software scope [1], developed after communication with stakeholders [1]
- A set of use-cases (scenario-based description) [1], developed in collaboration with the stakeholders [1]

Question 3. **[8]**

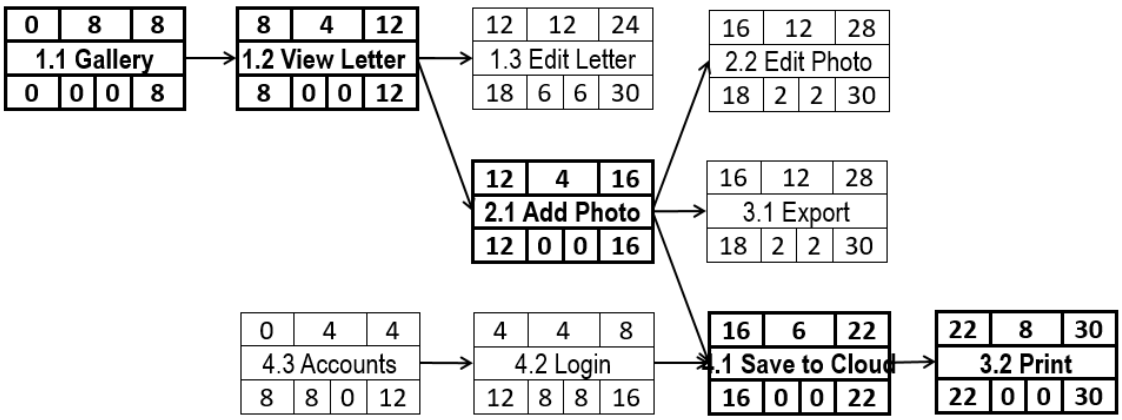
You have been asked to plan a project to build a mobile app to allow parents to collect photos corresponding to letters, to be composed into an alphabet book that can be delivered as an ebook or sent to a print shop to be printed as a personalized physical book. Given the following list of tasks, draw the network analysis diagram, and highlight the critical path.

#	Activity	Description	Length	Dependencies
1.1	Gallery	Show the pictures taken already, or show placeholders for letters which do not yet have photos	8	
1.2	View Letter	Show photo selected for a given letter	4	1.1
1.3	Edit Letter	Allow user to change the font and size of the letter, and specify placement on page	12	1.2
2.1	Add photo	Allow the user to select a photo from the phone's gallery for a particular letter	4	1.2
2.2	Edit photo	Allow the user to manipulate the photo (rotate, crop, filters	12	2.1
3.1	Export	Allow the user to export the photos to EPUB or PDF	12	2.1
3.2	Print	Allow the user to export the photos to a linked vendor's website for professional printing	8	4.1
4.1	Save to Cloud	Save photos and edits to the cloud	6	4.2, 2.1
4.2	Login	Allow entering of username and password to access cloud services	4	4.3
4.3	Accounts	Implement accounts	4	

For your reference:



== ANSWER ==



[2] Dependencies: 2 points total, minus half point for each error

[2] Critical Path: 2 points for correct identification of critical path, 1 point for partially correct

Check the following numbers

- [2] Earliest Finish: 1 point for getting 3.2 correct, ½ for 2.2, ½ for 3.1
- [1] Total float: Latest Start – Earliest Start (1 point for correct calculations, even if the numbers for LS and ES are incorrect)
- [1] Free float: ES (next) – EF (current) (l point for correct calculations)

Question 4. **[7]**

i. Explain what is meant by "low coupling" [2]

== ANSWER ==

Coupling is a measure of how strongly one element is connected to or has knowledge of other elements. High coupling results in the need to change large amounts of code when a small change is introduced. Thus software architecture should be designed such that coupling is low.

ii. Explain the purpose of the "creator" pattern [2]

== ANSWER ==

Problem: Who should be responsible for creating a new instance of some class?

Solution: Assign class C the responsibility to create an instance of class X if one or more of the following is true:

- C aggregates X objects.
- C contains X objects.
- C records instances of X objects.
- C closely uses X.
- C has the initializing data that will be passed to X when it is created (thus C is an Expert with respect to creating X).

iii. Explain the pros and cons of the Model-View-Controller architecture [3]

== ANSWER ==

- MVC allows the data to change independently of its representation and vice versa
- It is very easy to implement multiple presentations of the data (e.g. web, mobile) by implementing a new view without changing the model or the controller. Changes in one representation is automatically reflected in all views
- Can involve additional code and code complexity when the data model and interactions are simple.